

Business Requirements Document (BRD)

Sales & Customer Performance Analysis using SQL

Document Title	Business Requirements Document – Sales & Customer Performance Analysis
Project Name	Sales & Customer Performance Analysis using SQL
Prepared For	Business Stakeholders / Analytics & Sales Teams
Prepared Using	SQL (CTEs, Window Functions, Joins, Aggregations)
Document Status	Final

1. Introduction / Project Overview

This Business Requirements Document (BRD) defines the scope, objectives, data requirements, and analytical approach for the **Sales & Customer Performance Analysis** project. The project uses SQL to analyze sales transaction data in order to understand customer performance across different cities and across the years 2023 and 2024. The analysis is intended to help business stakeholders identify high-value customers, track revenue trends, and support data-driven decision-making for sales and customer relationship strategies.

2. Business Objective

The primary business objective of this project is to use SQL-based analysis to:

- Identify the top revenue-generating customers within each city.
- Analyze and compare customer performance across the years 2023 and 2024.
- Detect customers who consistently perform well over multiple years.
- Measure the growth or decline in customer revenue year-over-year.

These objectives directly support the business need to prioritize high-value customer relationships, allocate sales and marketing resources efficiently, and proactively respond to revenue decline patterns.

3. Problem Statement

The business currently lacks a structured, repeatable method to evaluate which customers contribute the most revenue in each city and whether that performance is sustained over time. Without this visibility, the organization cannot easily identify which customer relationships to prioritize, which are declining and require attention, or which cities are underperforming. This project addresses that gap

using SQL-driven analysis of the following objectives:

- Identify top revenue-generating customers in each city.
- Analyze customer performance across 2023 and 2024.
- Find customers who consistently perform well over time.
- Measure growth or decline in customer revenue.

4. Project Scope

4.1 In Scope

- Aggregation of sales data at the customer, city, and year level.
- Ranking of customers within each city using SQL window functions (RANK).
- Identification of the top 3 customers per city, per year.
- Year-over-year (2023 vs 2024) comparison of customer revenue performance.
- Classification of customer revenue trend as growth, decline, or no change.
- Insights on revenue concentration, consistency of top performers, and city-wise variation.

4.2 Out of Scope

- Predictive/forecasting models for future revenue (e.g., machine learning-based forecasting).
- Real-time or automated dashboard/reporting integration (analysis is SQL/query-based).
- Product-category-level profitability or margin analysis beyond category tagging via the Products\$ table.
- Customer demographic or behavioral segmentation beyond city and revenue.

5. Data Requirements / Dataset Used

The analysis relies on the following three source tables:

Table Name	Key Fields	Purpose
Customers\$	customer_id, city, segment	Provides customer identity, city location, and customer segment; used to group and rank customers geographically and by segment.
Orders\$	order_id, customer_id, product_id, order_date, sales	Provides transactional order-level sales data, used to compute revenue by customer, city, product, and year.
Products\$	product_id, category, product_name	Provides product-level context (category and name), used to enrich and contextualize order/sales data.

Data Relationships: Customers\$ and Orders\$ are joined on *customer_id* to compute customer-, city-, and segment-level revenue. Orders\$ is joined with Products\$ on *product_id* where product-level context (category/product name) is required for deeper analysis. The *order_id* field uniquely identifies each transaction, and *segment* allows customer performance to optionally be viewed by business segment in addition to city.

5.1 Data Profiling Summary (Actual Source Files)

Table	Row Count	Distinct Values / Notes
Customers\$	100 customers	10 cities (Delhi, Mumbai, Bangalore, Hyderabad, Chennai, Kolkata, Pune, Ahmedabad, Jaipur, Lucknow); 3 segments (Consumer, Corporate, Home Office). No missing values.
Orders\$	300 orders	All customer_id and product_id values reconcile fully against Customers\$ and Products\$ (no orphan records). Sales values range from approximately 307 to 49,995 per order. No missing values.
Products\$	30 products	3 categories (Electronics, Furniture, Clothing). No missing values.

Critical Data Quality Finding — order_date field: Profiling of the actual orders.csv file found that the order_date column does not contain valid calendar dates. Every one of the 300 order records currently holds the same placeholder-like value ('00:00.0') rather than a real date (e.g. DD-MM-YYYY). As a result, **no year value can currently be derived from this field**, which directly affects the core requirement to analyze and compare customer performance across 2023 and 2024. This is flagged as a high-priority data quality issue in Section 10 (Risks) and must be resolved — by correcting the source export or re-supplying a valid order_date column — before year-over-year analysis (FR-4, FR-5, FR-6, FR-7) can be executed against this dataset.

6. Functional Requirements

ID	Requirement	Description
FR-1	Aggregate Sales Data	Aggregate total sales at the customer, city, and year level from the Orders\$ table.
FR-2	Rank Customers by Revenue	Apply the SQL RANK() window function to rank customers by total revenue within each city.
FR-3	Identify Top Performers	Filter results to return the top 3 customers per city for each analysis year (2023 and 2024).
FR-4	Year-over-Year Comparison	Use INNER JOIN and/or LAG() to compare each customer's revenue between 2023 and 2024.
FR-5	Trend Classification	Use CASE WHEN logic to classify each customer's year-over-year performance as Growth, Decline, or No Change.
FR-6	Consistent Performer Identification	Identify customers who rank among the top performers in both 2023 and 2024.
FR-7	Revenue Difference Calculation	Calculate the absolute and/or percentage difference in revenue between the two years per customer.
FR-8	Product & Category Enrichment	Join Orders\$ with Products\$ on product_id to enable revenue analysis by product category where required.

ID	Requirement	Description
FR-9	Segment-Level Context	Incorporate the segment field from Customers\$ to allow customer performance to optionally be viewed by segment alongside city.

7. Analytical Approach / Methodology

The analysis follows a structured SQL-based methodology:

- **Step 1 — Data Aggregation:** Aggregate sales at the customer, city, and year level using GROUP BY.
- **Step 2 — Ranking:** Use window functions (RANK) partitioned by city and year to rank customers by revenue.
- **Step 3 — Filtering Top Performers:** Filter ranked results to retain only the top 3 customers per city per year.
- **Step 4 — Year Comparison:** Use INNER JOIN (or LAG) to bring 2023 and 2024 revenue for each customer into a single row for comparison.
- **Step 5 — Trend Classification:** Apply CASE WHEN logic to classify each customer's trend as revenue growth, decline, or stable performance.
- **Step 6 — Insight Generation:** Summarize findings on revenue concentration, consistency of top performers across years, and city-level variation.

Where deeper context is needed, Orders\$ is joined with Products\$ on *product_id* to bring in category and product-name detail, and with Customers\$ on *customer_id* to bring in city and segment detail, so that revenue trends can be viewed by city, by segment, or by product category.

Key SQL Concepts Used

- GROUP BY aggregations — for computing customer/city/year-level revenue totals.
- Common Table Expressions (CTEs) — for structuring multi-step queries in readable stages.
- Window Functions (RANK, LAG) — for ranking customers and comparing performance across years.
- INNER JOIN — for combining 2023 and 2024 performance data for comparison.
- CASE WHEN — for classifying revenue trends (growth/decline/no change).

8. Key Insights & Business Value

- A small group of customers contributes a disproportionately large share of total revenue.
- Certain customers consistently remain top performers across both 2023 and 2024.
- Revenue patterns and customer concentration vary significantly from one city to another.
- SQL-based ranking techniques provide an effective, repeatable way to track customer performance over time.

These insights enable the business to prioritize retention efforts for high-value and consistently top-performing customers, tailor city-specific sales strategies, and quickly flag customers with declining revenue for proactive account management.

9. Stakeholders

Stakeholder	Interest / Role
Sales Leadership	Uses insights to prioritize accounts and set city-wise sales targets.
Customer Success / Account Management	Uses top-performer and decline insights to manage key accounts.
Business/Data Analysts	Builds, maintains, and extends the SQL analysis.
Marketing Team	Uses customer and city trends to target campaigns and offers.

10. Risks & Data Quality Issues

ID	Risk / Issue	Impact	Recommended Action
R-1	order_date field in Orders\$ contains no valid dates (all 300 rows = '00:00.0')	High — year cannot currently be derived, blocking all 2023 vs 2024 comparison requirements (FR-4 to FR-7)	Request a corrected orders export with a valid order_date (DD-MM-YYYY or similar) from the source system before proceeding with year-over-year analysis.
R-2	Small dataset size (100 customers, 300 orders, 30 products)	Medium — top-3-per-city rankings and trend results may be less statistically robust and more sensitive to individual outlier orders	Treat current results as a proof-of-concept / pilot; revalidate logic once full-scale production data is available.
R-3	No explicit year/date dimension elsewhere in the schema to cross-check order_date	Medium — cannot independently validate or reconstruct the missing year information from other fields	Confirm with the data source owner whether a separate timestamp, fiscal period, or log field exists that can substitute for order_date.

11. Assumptions & Constraints

Assumptions

- The Orders\$ table is expected to contain accurate order-level sales data (order_id, customer_id, product_id, order_date, sales) with a valid, resolvable order_date for the years 2023 and 2024 — see Risk R-1 regarding the current state of this field.
- Each customer_id in Orders\$ has a corresponding, unique entry in Customers\$ with a valid city and segment value (confirmed true in the profiled source files — no orphan records found).
- Each product_id in Orders\$ has a corresponding, unique entry in Products\$ with valid category and product_name values (confirmed true in the profiled source files — no orphan records found).
- "Revenue" is derived directly from the sales field in the Orders\$ table without additional cost/discount adjustments unless specified.

Constraints

- Analysis is limited to the data available in Customers\$, Orders\$, and Products\$ tables (100 customers, 300 orders, 30 products in the currently supplied files).
- The comparison is intended to cover two years (2023 and 2024); this is currently blocked pending resolution of the order_date data quality issue (Risk R-1).
- Analysis is conducted via SQL queries; no dashboard/BI tool integration is included in this scope.

12. Expected Outcome / Business Outcome

This project delivers a repeatable SQL-based analytical framework that helps the business understand customer behavior, identify high-value and consistently performing customers, and track revenue growth or decline across cities and years. The outcome supports better-informed, data-driven decisions in sales strategy, customer relationship management, and resource allocation.